Muhammad Ali

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Education

William & Mary Williamsburg, VA
Bachelor of Science in Computational & Applied Mathematics & Statistics (CAMS) *Expected May 2027*

Selected Coursework

Mathematics

MATH 451/551 – Probability
MATH 452/552 – Mathematical Statistics
MATH 455/555 – Statistical Learning
MATH 311 – Real Analysis
MATH 403 – Measure Theory *(planned)*
MATH 405 – Complex Analysis

Computer Science

CSCI 416/516 – Fundamentals of AI/ML
CSCI 436/536 – Data Mining *(planned)*
CSCI 446/546 – Neural Networks for ML
CSCI 456/556 – Large Language Models *(planned)*
CSCI 420/520 – High-Dimensional Geometry of Data
CSCI 435/535 – Software Engineering

Projects

TDA Time-Series Embeddings | *Python, Jupyter, Topological Data Analysis* In Progress

- Exploring delay embeddings, point-cloud geometry, persistent homology, and visualization for periodic time-series data.
- Developing reproducible computational notebooks connecting geometric intuition with topology-inspired analysis.
- Preparing project materials for portfolio presentation, including mathematical motivation, code, and visual outputs.

Topological Structure in Software Systems | *Java, Python, Graphs, TDA* Planned

- Planned research-style project applying topological data analysis to Java source code structure.
- Project design involves abstract syntax trees, graph representations, persistent homology, and structural complexity metrics.
- Intended to study patterns in software architecture through computational and topology-inspired representations.

Neural Network Representation Geometry | *Python, Neural Networks, Embeddings* Planned

- Planned project implementing a small neural language model inspired by early neural probabilistic language modeling.
- Project design includes word embeddings, next-word prediction, cross-entropy loss, and representation analysis.
- Intended to connect neural network implementation with geometric interpretation of learned representations.

Technical Skills

Languages: Python, R, C++, LaTeX
Tools: Git, GitHub, Jupyter Notebook, VS Code
Libraries: NumPy, pandas, Matplotlib, scikit-learn

Additional Experience

Prospective Teaching Assistant / Grader, MATH 451/551 & MATH 452/552 2026–2027
William & Mary Department of Mathematics *Williamsburg, VA*

- Anticipated role supporting MATH 451/551 Probability and MATH 452/552 Mathematical Statistics through grading and academic course support.
- Preparation emphasizes probability theory, statistical inference, mathematical communication, and proof-based problem solving.